

Common Biology File Formats

Working with Biological Data in Python and R · what each format stores and when you'll meet it

1 • Raw & Aligned Sequence Data

Format (extension)	What it stores	Commonly used for	Type & tools
FASTA .fasta .fa .fna .faa	Nucleotide or protein sequences: a ">" header line followed by the sequence.	Reference genomes, gene & protein sequences, BLAST databases.	<i>Plain text</i>
FASTQ .fastq .fq .fq.gz	Sequencing reads plus a per-base quality score for every base (4 lines/read).	Raw output of sequencers (Illumina, etc.); the start of most pipelines.	<i>Plain text, usually gzipped</i>
SAM .sam	Reads aligned to a reference genome, with position, quality, and flags.	Output of aligners (BWA, Bowtie2, STAR, minimap2).	<i>Plain text, tab-delimited; samtools</i>
BAM .bam	The compressed binary form of SAM — same content, much smaller.	Storing & indexing alignments; the standard for mapped reads.	<i>Binary; samtools, IGV</i>
CRAM .cram	Reference-based compressed alignments (stores only differences vs the reference).	Long-term archival of alignments; smaller than BAM.	<i>Binary; needs the reference</i>

2 • Genome Annotation & Intervals

Format (extension)	What it stores	Commonly used for	Type & tools
GFF / GFF3 .gff .gff3	Feature annotations (genes, exons, CDS) with genomic coordinates, in 9 columns.	Describing where features sit on a genome; genome browsers.	<i>Plain text, tab-delimited</i>
GTF .gtf	A GFF variant focused on gene/transcript models.	Gene models for RNA-seq quantification (featureCounts, HTSeq, StringTie).	<i>Plain text</i>
BED .bed	Genomic intervals: chromosome, start, end (0-based), + optional fields.	Regions of interest: peaks, exons, primers, target panels.	<i>Plain text; bedtools</i>
GenBank .gb .gbk	An annotated sequence with rich metadata (features, references, organism).	NCBI sequence records, plasmid maps, single-gene annotation.	<i>Plain text</i>
WIG / bigWig .wig .bw	A continuous numeric signal along the genome (e.g. read coverage).	Coverage & ChIP/ATAC signal tracks in genome browsers.	<i>Text / binary; UCSC tools</i>
narrowPeak .narrowPeak	BED-like peak calls with significance and summit position.	ChIP-seq / ATAC-seq peak output (MACS2).	<i>Plain text</i>

3 • Genetic Variants

Format (extension)	What it stores	Commonly used for	Type & tools
VCF .vcf	Variants (SNPs, indels) relative to a reference, plus per-sample genotypes.	Variant calling & genotyping output (GATK, bcftools); population genetics.	<i>Plain text; bcftools</i>
BCF .bcf	The compressed binary form of VCF.	Large cohorts / many-sample variant data.	<i>Binary; bcftools</i>
gVCF .g.vcf	A VCF recording every position (variant and non-variant) with confidence.	Joint genotyping across many samples.	<i>Text or binary; GATK</i>

4 • Expression & Single-Cell Data

Format (extension)	What it stores	Commonly used for	Type & tools
Count matrix .csv .tsv	A table of genes (rows) × samples (columns) of read counts.	Bulk RNA-seq differential expression input (DESeq2, edgeR).	<i>Plain text</i>
Matrix Market .mtx	A sparse matrix (mostly zeros), with separate barcodes & features files.	Single-cell count matrices (10x Genomics output).	<i>Plain text trio</i>
AnnData .h5ad	A matrix plus aligned cell and gene metadata in one container.	Single-cell analysis in Python (Scanpy).	<i>Binary (HDF5)</i>
Loom .loom	Large omics matrices with row/column attributes.	Single-cell & large expression datasets.	<i>Binary (HDF5)</i>
CEL .cel	Raw probe intensities from a microarray chip.	Affymetrix microarray expression.	<i>Binary</i>

5 • Multiple Alignments & Phylogenetic Trees

Format (extension)	What it stores	Commonly used for	Type & tools
Clustal / FASTA-aln .aln .fasta	A multiple sequence alignment (sequences padded with gaps).	Viewing & sharing alignments (Clustal, MUSCLE, MAFFT output).	<i>Plain text</i>
Stockholm .sto .stk	An alignment plus per-column / per-family annotation.	Profile & family alignments (Pfam, Rfam, HMMER).	<i>Plain text</i>
PHYLIP .phy	A compact alignment format for phylogenetics programs.	Input to tree-building tools.	<i>Plain text</i>
Newick .nwk .tree .nh	A tree's topology with branch lengths and support, nested in parentheses.	Phylogenetic trees (ape, ete3, FigTree).	<i>Plain text</i>
NEXUS .nex	A richer container: trees plus the alignment, plus analysis commands.	Bayesian phylogenetics (MrBayes, BEAST).	<i>Plain text</i>

6 • Macromolecular & Chemical Structures

Format (extension)	What it stores	Commonly used for	Type & tools
PDB .pdb	3-D atomic coordinates of proteins / nucleic acids (legacy format).	Protein & nucleic-acid structures (PyMOL, Chimera).	<i>Plain text (fixed columns)</i>
mmCIF / PDBx .cif	The modern structure format; handles very large complexes & metadata.	The current Protein Data Bank standard.	<i>Plain text</i>
MOL / SDF .mol .sdf	Small-molecule structures (atoms, bonds), one or many per file.	Ligands, compounds, cheminformatics (RDKit).	<i>Plain text</i>

7 • Mass Spectrometry & Proteomics

Format (extension)	What it stores	Commonly used for	Type & tools
mzML .mzml	Raw mass-spectrometry spectra in an open standard.	Proteomics / metabolomics raw data exchange.	<i>XML text</i>
mzIdentML / mzTab .mzid .mzTab	Peptide/protein identifications and quantifications.	Reporting MS search results.	<i>XML / tab text</i>
FASTA (protein) .fasta .faa	Protein sequences used as a search database.	Database search (UniProt, MaxQuant, MSGF+).	<i>Plain text</i>

8 • General-Purpose & Container Formats

Format (extension)	What it stores	Commonly used for	Type & tools
CSV / TSV .csv .tsv	Any rectangular table — the universal lingua franca of data.	Metadata, sample sheets, count tables, results.	<i>Plain text; read_csv / read.csv</i>
Excel .xlsx .xls	Spreadsheets with multiple sheets and formatting.	Small datasets, sample sheets shared by collaborators.	<i>Binary (zipped XML)</i>
HDF5 .h5 .hdf5	Hierarchical, compressed numeric arrays in one file.	Large numeric datasets (underlies h5ad, loom).	<i>Binary</i>
JSON / YAML .json .yaml	Nested, structured key-value data.	Pipeline configs, API responses, metadata.	<i>Plain text</i>
Parquet .parquet	Columnar, compressed tabular storage for big data.	Large dataframes; fast analytic queries.	<i>Binary; pandas/Arrow</i>

9 • Index & Reference Helper Files

Format (extension)	What it stores	Commonly used for	Type & tools
FASTA index .fai	A small index of a FASTA file's record positions.	Fast random access to a reference (samtools faidx).	<i>Plain text</i>
Sequence dictionary .dict	A catalogue of the reference's contigs and lengths.	Required reference companion for GATK.	<i>Plain text</i>
Alignment / variant index .bai .csi .tbi	An index over a sorted BAM or bgzipped VCF/GFF.	Jumping to a genomic region without reading the whole file.	<i>Binary; samtools / tabix</i>

Rules of thumb: *text formats (FASTA, VCF, GFF, BED, Newick, CSV) are human-readable and easy to peek at with head; binary formats (BAM, BCF, h5ad, HDF5) are smaller and faster but need a tool to open. A .gz suffix just means the file is gzip-compressed — most tools read it directly. When in doubt, open a text file with `head -n 5 file` to see its structure.*